Practice Questions: Lists, Tuples and Dictionaries

Ten practice problems in the style of Contest-11 (same difficulty, different questions).

Rules

No built-in functions or methods are allowed. Allowed only: `input()`, `print()`, `int()`, `.split()`, `range()`, loops, if/else, operators, list indexing and slicing, and dictionary indexing with the `in` operator. For tuples, creating them with `( )`, indexing and unpacking are assumed to be fine, since they are not in the banned list.

Lists

Q1 | List | Easy: The Conveyor Belt

A conveyor belt carries n parcels with weights w1…wn. The belt rotates to the right k times. Each rotation moves the last parcel to the front. Print the arrangement after k rotations.

- **Input:** Line 1: `n k`. Line 2: n integers.
- **Output:** n space-separated integers.
- **Constraints:** 1 ⩽ n ⩽ 1000, 0 ⩽ k ⩽ 10^9 (k can be larger than n).

Example

Input:

```
5 2
1 2 3 4 5
```

Output:

```
4 5 1 2 3
```

Q2 | List | Medium: Silver Medal

n athletes have scores. Find the second-highest distinct score, meaning the largest score strictly smaller than the maximum. If there is no such score, print -1.

- **Input:** Line 1: n. Line 2: n integers.
- **Output:** One integer.

- **Constraints:** 1 ⩽ n ⩽ 1000, 0 ⩽ score ⩽ 10^9.

Example 1

Input:

```
6
4 9 9 2 7 7
```

Output:

```
7
```

Example 2

Input:

```
3
5 5 5
```

Output:

```
-1
```

Q3 | List | Easy–Medium: The Guest List

A registration desk recorded n guest IDs, and some guests registered more than once. Print the IDs with duplicates removed, keeping only the first occurrence of each ID and preserving the original order.

- **Input:** Line 1: n. Line 2: n integers.
- **Output:** The unique IDs, space-separated.
- **Constraints:** 1 ⩽ n ⩽ 1000, 1 ⩽ ID ⩽ 10^6.

Example

Input:

```
7
4 2 4 3 2 5 3
```

Output:

```
4 2 3 5
```

Q4 | List + Sorting | Medium: Two Queues, One Counter

Two queues of customers are already sorted by token number, in ascending order. Merge them into one sorted queue without using any sort. Write the merging logic yourself.

- **Input:** Line 1: `n m`. Line 2: n integers. Line 3: m integers.
- **Output:** n+m sorted integers.
- **Constraints:** 1 ⩽ n, m ⩽ 1000, values up to 10^9, duplicates allowed.

Example

Input:

```
3 4
1 4 9
2 3 4 10
```

Output:

```
1 2 3 4 4 9 10
```

Q5 | List | Medium: Gift Pairs

Meera wants to buy two different gifts from a shop with n gifts so that their prices add up exactly to her budget T. Count the pairs of positions (i, j) with i < j whose prices sum to T.

- **Input:** Line 1: `n T`. Line 2: n integers (prices).
- **Output:** One integer, the number of pairs.
- **Constraints:** 2 ⩽ n ⩽ 1000, 1 ⩽ price ⩽ 10^6.

Example

Input:

```
5 6
1 5 3 3 4
```

Output:

```
2
```

Explanation: the pairs are 1+5 and 3+3.

Tuples

Q6 | Tuple | Easy: The Nearest Tower

A drone is at the origin (0, 0). There are n signal towers, each at coordinates (x, y), stored as a tuple. Print the coordinates of the tower closest to the origin, using squared distance $x^2 + y^2$. If there is a tie, print the one that appears first.

- **Input:** Line 1: n. Next n lines: `x y`.
- **Output:** `x y` of the closest tower.
- **Constraints:** $1 \leqslant n \leqslant 1000$, $-1000 \leqslant x, y \leqslant 1000$.

Example

Input:

```
4
3 4
-1 2
0 -2
5 5
```

Output:

```
0 -2
```

Q7 | Tuple + Sorting | Medium: The Leaderboard

Store each student as a tuple (name, marks). Print all students from highest to lowest marks. Students with equal marks must stay in their original input order. Write the sort yourself, and make sure it is stable.

- **Input:** Line 1: n. Next n lines: `name marks`.
- **Output:** n lines, each `name marks`.
- **Constraints:** $1 \leqslant n \leqslant 1000$, names are lowercase words, $0 \leqslant$ marks $\leqslant 100$.

Example

Input:

```
4
amit 70
bina 85
chetan 70
dev 90
```

Output:

```
dev 90
bina 85
amit 70
chetan 70
```

Dictionaries

Q8 | Dictionary | Medium: The Popular Word

A blogger wants to know which word she used the most. Given n words, print the most frequent word and its count. If several words tie, print the one that appeared first in the input.

- **Input:** Line 1: n. Line 2: n lowercase words.
- **Output:** `word count`.
- **Constraints:** 1 ⩽ n ⩽ 10^5, word length 1 to 15.

Example

Input:

```
7
apple ball apple cat ball apple cat
```

Output:

```
apple 3
```

Q9 | Dictionary | Medium: The Corner Store

A store starts with n items and their stock. Then q transactions arrive, each either `add item qty` or `sell item qty`.

- `add` increases the stock. If the item is new, it is created.
- `sell` decreases the stock. If the item doesn't exist or the stock is too low, print `Insufficient stock` immediately and ignore that transaction.

After all transactions, print every item with its stock in the order the items first appeared.

- **Input:** Line 1: `n q`. Next n lines: `item stock`. Next q lines: a transaction.
- **Output:** Any `Insufficient stock` lines first, then the final `item stock` lines.
- **Constraints:** 1 ⩽ n, q ⩽ 10^5, 0 ⩽ qty ⩽ 10^6.

Example

Input:

```
2 3
pen 10
book 5
sell pen 4
sell book 6
add book 3
```

Output:

```
Insufficient stock
pen 6
book 8
```

Q10 | Dictionary | Easy: Scrambled Letters

Two words are anagrams if one can be made from the other by rearranging letters, using every letter exactly once. Using dictionaries to count letters, check whether two given words are anagrams. Print `Yes` or `No`.

- **Input:** Two lines, one word each (lowercase letters only).
- **Output:** `Yes` or `No`.
- **Constraints:** 1 ⩽ word length ⩽ 10^5.

Example 1

Input:

```
listen
silent
```

Output:

```
Yes
```

Example 2

Input:

```
hello
world
```

Output:

```
No
```

Next steps

Test cases (including edge cases) and model solutions in the same no-built-in style can be written for any of these. For harder practice, Hard-level questions such as grouping anagrams or Two Sum with a dictionary can be added.